

VM design considerations (runner + runtime per environment)

VM design considerations (client-specific)

Context: ENGAGEMENT-ALIGNMENT.md — **SOW items 2 and 4** (standards + environment improvement). **No prod Linux today; dev = Podman**; target path **dev → UAT → prod**. This outline supports **UAT/prod Linux VM design** and **dev host remediation**. **Client builds** VMs; consultant advises.

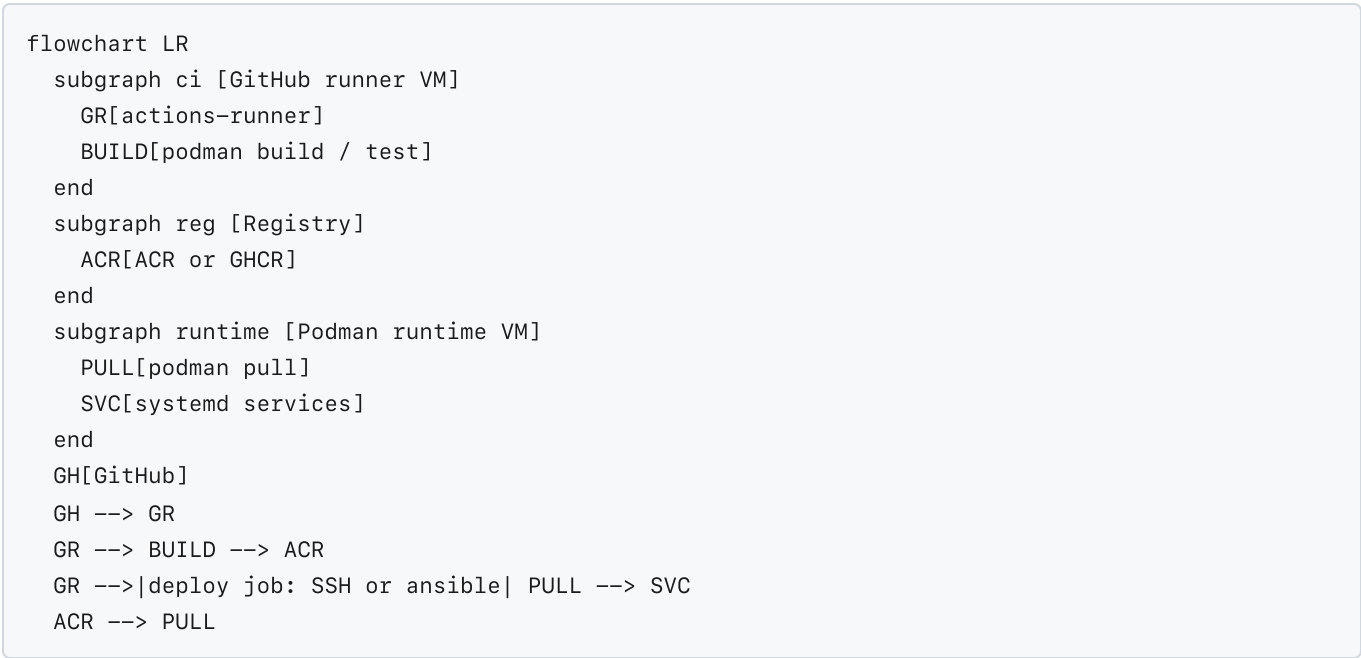
Related: discovery-workshop-2.md · ARCHITECTURE.md · AZURE-NETWORK-ARCHITECTURE.md · reference module `infra/terraform/modules/compute-linux/`

1. VM role and environment

1.1 Recommended topology: split GitHub Actions runner from Podman runtime

Decision: Use **two VMs per environment** (dev / UAT / prod)—do **not** co-locate the **GitHub self-hosted runner** with Podman workload hosts.

VM role	Purpose	Must not run
Podman runtime VM	Long-lived services: FastAPI microservices, Dagster, ingestion staging, <code>systemd /quadlet</code>	<code>actions-runner</code> , <code>podman build</code> , compile toolchains, CI job execution
GitHub runner VM	CI/CD only: clone, test, build image , push to registry, trigger deploy to runtime VM	Application containers, Dagster daemon, ingestion data paths



Why split

Risk on combined host	Split mitigates
CI spikes steal CPU/RAM from APIs and Dagster	Independent SKU sizing
Compromise of runner (untrusted job code) exposes runtime secrets	Blast radius isolation
podman build + dev packages on prod runtime	Runtime stays pull-only
Pen test / hardening scope blurred	Runner = build zone; runtime = production zone

1.2 Environment matrix

Consideration	Dev (today)	UAT (next)	Prod (first Linux)
Podman runtime VM	Stabilize existing services; remove runner when split VM exists	Prod-like runtime; validate deploy from runner	Snowflake data platform runtime, APIs, orchestration
GitHub runner VM	Document current runner placement; plan migration off runtime host	Same runner pattern as prod; may share dev runner initially	Dedicated runner; no app workloads
Subscription	Dev tenant/sub	Dev or dedicated UAT RG	Prod subscription (separate from dev)
Parity	Source of truth to document	Two VMs + same Podman + systemd deploy as prod	Same images and deploy mechanism as UAT
Count	1+ hosts today (often combined)	2 VMs minimum per env	2 VMs minimum; scale runtime only if workload grows

Still open: Single runtime VM vs split ingestion vs API on **runtime** tier only—runner stays separate either way.

2. Workload profile (drives size & disk)

From discovery, the VM must support:

Workload	VM	Implication
Podman containers (FastAPI / microservices)	Runtime	CPU + RAM for concurrent services; systemd or quadlet per service
Dagster orchestration	Runtime	CPU spikes during runs; confirm co- location with APIs on same runtime host
GitHub Actions runner (CI jobs)	Runner only	Size for parallel jobs + build cache; bursty CPU; never on runtime VM
Image build (podman build)	Runner only	Build tools, caches, test deps live here —not on UAT/prod runtime
SFTP / file ingestion → Snowflake path	Runtime	Data disk for staging; IOPS for file churn
Snowflake / external DB connectivity	Egress heavy; no assumption of private link unless client provides	
Charles River / trading APIs	Low-latency egress; document endpoints for firewall allow-list	
Retool / Power BI	Mostly SaaS—VM may not host; verify no UI on VM	

Sizing: Start assessment from **current dev utilization** (`top` , `podman stats` , disk usage) before UAT/prod SKU.

3. Operating system

Item	Recommendation	Client action
OS	RHEL 9.x (align with dev and RH support)	RHSM / Satellite
Container runtime	Podman (not Docker)—client standard	podman , podman–compose or quadlet
Init	systemd for service lifecycle + reboot persistence	Unit files in git
Package baseline	Minimal + audit, chrony, rsyslog, sssd (if Entra/AD)	Standards doc
Marketplace image	RHEL BYOS or golden SIG if fleet grows	Licensing
Patching	dnf update --security ; maintenance windows	Client ops

Avoid: Mixing Docker and Podman on same host without reason.

4. Compute (SKU)

Size **each role independently** from dev metrics.

Podman runtime VM

Factor	Guidance
vCPU	Concurrent Podman services + Dagster peaks only— exclude CI
RAM	Container sum + OS + Dagster; add 25–30% headroom
SKU family	Dsv5 or Esv5; E if large in-memory work on VM
Burstable (B-series)	Avoid for UAT/prod runtime

GitHub runner VM

Factor	Guidance
vCPU	Peak concurrent jobs × cores per job (often 2–4 per job); measure from current runner
RAM	Largest build/test job + parallel job count; builds are often RAM-bound
Disk	Larger OS disk or ephemeral cache for layer/build artifacts; prune policy required
SKU family	Dsv5 (CPU-heavy builds) or Fsv2 if compile-heavy; still avoid B-series for prod runner
Count	One runner VM per env to start; add second runner VM only if job queue SLA requires

Shared	Guidance
Accelerated networking	Enable if supported and traffic warrants
Gen2	Required for Trusted Launch

Action: Export **runtime** and **runner** metrics separately for 2 weeks before locking UAT/prod SKUs.

5. Storage (critical — prior mount failures)

Use **both** patterns where needed: **Managed Disk** (VM-local block) + optional **Azure File Share** (shared files).

Type	Azure service	Reference module	Primary use
Managed Disk	Managed Disk attached to VM	<code>compute-linux</code> <code>data_disks</code>	Podman volumes, local ingestion staging, high IOPS
File Share	Azure Files (NFS or SMB)	<code>storage-fileshare</code>	Shared ingestion drop, multi-consumer files, hybrid access

Podman runtime VM

Disk / share	Purpose	Considerations
OS disk	<code>/</code> , <code>/var</code> (unless data disk used for Podman)	Premium SSD for prod; 64–128 GB minimum
Managed data disk	Ingestion staging, app data, Podman volumes	Premium_LRS ; separate LUN; mount via UUID in fstab or cloud-init — <code>compute-linux data_disks</code>
Azure File Share (optional)	Shared landing zone when multiple systems need same files	NFS on RHEL (Premium FileStorage); private endpoint; runtime UAMI Storage File Data Privileged Contributor; mount at layer 2 via <code>/opt/compliance/bootstrap/mount-azure-fileshare.sh</code>

Default: managed data disk for runtime. Add file share when discovery confirms shared ingestion or off-VM producers.

GitHub runner VM

Disk	Purpose	Considerations
OS disk	Runner install, job workspaces, build cache	128–256 GB typical; aggressive cleanup (<code>podman system prune</code> , Actions workspace cleanup)
Managed data disk	Optional	Build cache or large fixtures only — not app ingestion
File Share	Usually not needed	Runner uses local disk + registry

Shared (both VMs)

Item	Considerations
Persistence	Survive reboot— lesson from dev: improper mount caused data loss; automate in IaC/cloud-init
Encryption	Managed Disk + Files: Azure SSE at rest; CMK if policy requires
Backup	Managed data disk in ASR/Backup; file share via share snapshot / backup policy
Network	File share: private endpoint to <code>privatelink.file.core.windows.net</code> ; egress from VM subnet

Anti-pattern	Fix
Manual mount after reboot	<code>fstab</code> + <code>systemd</code> mount units + cloud-init on first boot
Podman volumes on OS disk only	Dedicated managed data disk for <code>/var/lib/containers</code> or named volumes path
Storage account keys on VM	UAMI RBAC to file share; avoid account keys in <code>fstab</code>

Reference Terraform (runtime VM)

```
module "podman_runtime" {
  source = "../../modules/compute-linux"
  # ...
  data_disks = [{
    name           = "vm-prod-runtime-data01"
    size_gb        = 256
    storage_account_type = "Premium_LRS"
    lun            = 0
    mount_hint     = "/var/lib/containers"
  }]
}

module "ingestion_share" {
  source = "../../modules/storage-fileshare"
  # ...
  share_protocol = "NFS"
  principal_ids  = [azurerm_user_assigned_identity.runtime.principal_id]
}
```

6. Identity and access

6.1 Human access (both VMs)

Item	Dev issue	Target for UAT/prod
Shared admin	All developers one account	Per-user Linux accounts or Entra ID SSH (Patrick/Kirk pilot)
SSH	Direct login	Bastion or Entra SSH; no shared keys
Sudo	Uncontrolled	<code>sudo</code> via group; logged (auditd)
Break-glass	—	Documented; disabled by default

VM admin user: Dedicated Linux account for automation; no day-to-day human shared login.

6.2 Azure managed identity + Azure CLI (both VMs)

Azure has **no** built-in “VM role” that enables Azure CLI. **Both** the GitHub runner VM and the Podman runtime VM use the same platform pattern:

1. **User-assigned managed identity (UAMI)** attached to the VM at build (see `compute-linux` module `identity` block).
2. `azure-cli` installed in baseline (Ansible/cloud-init) on **both** hosts.
3. `az login --identity` (or `-u <uami-resource-id>` for user-assigned) — no client secret on disk.
4. **Azure RBAC** role assignments on each UAMI principal — **different roles per VM**, least privilege.

```

flowchart TB
    subgraph runner [GitHub runner VM]
        RMI[UAMI github-runner]
        RAZ[az login --identity]
        RMI --> RAZ
    end
    subgraph runtime [Podman runtime VM]
        PMI[UAMI podman-runtime]
        PAZ[az login --identity]
        PMI --> PAZ
    end
    subgraph azure [Azure RBAC]
        ACR[ACR]
        KV[Key Vault]
    end
    RAZ -->|AcrPush + KV read| ACR
    RAZ --> KV
    PAZ -->|AcrPull + KV read| ACR
    PAZ --> KV

```

Naming (per environment): e.g. `uami-{env}-github-runner`, `uami-{env}-podman-runtime` — **one UAMI per VM**; do not share a single identity across both roles.

Capability	GitHub runner UAMI	Podman runtime UAMI
Azure RBAC: AcrPush	Yes — push built images	No
Azure RBAC: AcrPull	Optional (if job pulls base images)	Yes — deploy-time pull
Azure RBAC: Key Vault Secrets User	Yes — deploy/CI secrets	Yes — app/runtime secrets
Azure RBAC: Storage Blob Data *	Only if pipeline reads/writes blobs	Only if ingestion uses blob
Contributor / broad RG write	No	No
az on host	Yes — registry login, KV, deploy automation	Yes — pull-only operations; avoid general ARM changes

On-VM usage (examples):

```
# Runner — after image build
az login --identity -u "$RUNNER_UAMI_ID"
az acr login --name <acr>
podman push <acr>.azurecr.io/app:${GIT_SHA}

# Runtime — deploy or systemd ExecStartPre
az login --identity -u "$RUNTIME_UAMI_ID"
az acr login --name <acr>
podman pull <acr>.azurecr.io/app:${TAG}
```

IMDS: Managed identity tokens use the instance metadata endpoint (169.254.169.254). Do **not** block this on the VM — it is local to the host.

CI/CD deploy: Prefer **managed identity** (runner UAMI or GitHub OIDC) over stored service principal secrets. Human SSH to prod for deploy is not the target state.

GitHub OIDC (complement to runner UAMI): Workflows may use `azure/login@v2` with federated credentials instead of relying solely on VM MI for every job — client chooses one primary pattern; runtime still uses **runtime UAMI** for pull.

Item	Owner
Create UAMI + attach to VM	Client platform (Terraform/Bicep)
RBAC role assignments per UAMI	Client platform / IAM
Install <code>azure-cli</code> + document UAMI IDs in cloud-init	Client platform + standards doc

7. Azure VM platform

Item	Recommendation
Trusted Launch	<code>secure_boot_enabled</code> , <code>vtm_enabled</code> (see <code>compute-linux</code> module)
Managed identity	User-assigned UAMI per VM — runner and runtime each get a dedicated identity (§6.2)
Patch mode	<code>ImageDefault</code> or <code>AutomaticByPlatform</code> per policy
Availability	Prod: consider Availability Zone if SLA requires
Proximity	Same region as Snowflake/Azure data services where possible
Tags	Environment, owner, cost center, compliance scope, <code>Role=github-runner</code> or <code>Role=podman-runtime</code>
Boot diagnostics	Enabled for break-fix

8. Networking

Podman runtime VM

Item	Consideration
Subnet	App/data tier; no public IP on prod if policy allows
NSG inbound	Bastion; APIM or internal consumers only— no runner SSH from internet
NSG egress	Snowflake, RHSM, ACR/GHCR (pull), SFTP peers, Charles River APIs; Azure control plane ; Azure Files private endpoint if share used
Private IP	Static if firewalls whitelist IP
APIM	Edge auth may front services—runtime accepts traffic from APIM subnet only

GitHub runner VM

Item	Consideration
Subnet	CI/build tier —separate subnet or NSG from runtime if policy allows
NSG inbound	Bastion; runner polls GitHub outbound-only (no inbound from GitHub required for standard self-hosted runner)
NSG egress	GitHub (<code>github.com</code> , <code>api.github.com</code> , <code>codeload.github.com</code> , <code>objects.githubusercontent.com</code>), registry push (ACR/GHCR), RHSM, package mirrors; Azure control plane (<code>login.microsoftonline.com</code> , <code>management.azure.com</code>) for runner UAMI / <code>az acr login</code>
To runtime VM	Deploy job: SSH (key in Key Vault), Ansible , or pull-only (runtime pulls; runner only signals restart)—document one pattern
Not required on runtime	GitHub API egress if runtime never runs CI

Shared	Consideration
DNS	Client resolver (on-prem AD forwarder or Azure DNS private resolver)
Hybrid	Head data center paths—document in network requirements
Network not in IaC	Client implements peering/NSG/firewall via tickets

Reference: NETWORK-DISCOVERY-QUESTIONNAIRE.md, AZURE-NETWORK-ARCHITECTURE.md.

9. Podman / container operations

Item	Consideration
Rootful vs rootless	Document dev choice; prod often rootful with SELinux policies or hardened rootless
SELinux	Mandatory policy for container paths (:z , :Z , custom booleans)—fix dev conflicts before replicate
Images	Registry (ACR vs GHCR)—no <code>podman build</code> on prod server
ACR auth (runtime)	Runtime UAMI + <code>az acr login --identity</code> or equivalent; <code>AcrPull</code> only
Deploy	CI pushes image → runtime pulls via UAMI → <code>systemd</code> restart; no manual deploy
Secrets	Key Vault + runtime UAMI (<code>Key Vault Secrets User</code>); not in git or world-readable env files
Logging	Container logs → <code>journald</code> (capped, rate-limited) → <code>rsyslog</code> ; <code>logrotate</code> + <code>/var/log/archive</code> ; <code>audit-forward</code> log for SIEM
Resource limits	<code>podman run --cpus --memory</code> or <code>systemd MemoryMax</code>

Walkthrough required: build → store → deploy (document current vs target).

10. Security and compliance

Item	Source	Action
Pen test / Arctic Wolf	Workshop 2 / Kirk	P0 closed on dev before UAT; re-scan before prod — see STANDARDS-RHEL-PODMAN-v0.1.md §5 gates
CIS RHEL 9 L1	Standards (primary)	OpenSCAP in AIB; verify artifacts
DISA STIG	GRC only if mandated	Separate OpenSCAP profile — not default; app compatibility test required
FIPS	Discussed	Only if apps approved— app team tests
auditd	Baseline	Identity, sudo, execve, Podman; <code>audisp syslog</code> + archive retention
dnf-automatic	Baseline	Security-only updates; reboot deferred to change window
AIDE / EDR	Bank norm	Per client SOC (Logic Monitor + EDR)
No secrets on disk	—	KV references only

11. CI/CD integration (GitHub Actions → registry → Podman runtime)

Step	Where	Pattern
1. Trigger	GitHub	Workflow on push/PR/tag; branch protection for UAT/prod
2. Test / build	Runner VM	Job runs-on: [self-hosted, rhel, uat]; workflow steps run podman build / tests on runner host
3. Push	Runner VM → registry	Tag app:git-sha; runner UAMI az acr login --identity + podman push; or GHCR via GITHUB_TOKEN
4. Deploy	Runner VM → Runtime VM	Preferred: trigger runtime pull via runtime UAMI (systemd/ ExecStartPre) or SSH/Ansible — runtime does not build
5. Promote	GitHub Environments	dev → UAT → prod; required reviewers on prod environment
Secrets	Key Vault + UAMI / GitHub OIDC	No SP passwords on either VM; Snowflake creds via KV + runtime UAMI

Item	Consideration
Runner registration	Org- or repo-level self-hosted runner; one runner VM per env with labels (dev , uat , prod)
Runner service	actions-runner under systemd; pin runner version in standards; auto-update policy per security
Runner UAMI	AcrPush, Key Vault Secrets User; azure-cli installed; document UAMI ID in /etc/environment or cloud-init
Runtime UAMI	AcrPull, Key Vault Secrets User only; azure-cli for az acr login — no AcrPush
GitHub OIDC	Optional azure/login@v2 in workflow (federated credential) — use instead of or with runner UAMI; runtime still uses runtime UAMI
Concurrency	Workflow concurrency: + job limits ≤ vCPU budget; avoid parallel builds starving deploy jobs
Untrusted code	Runner VM is lower trust — runner UAMI must not inherit runtime RBAC or Contributor
IaC	Client Terraform/Bicep: two UAMIs , two VMs, RBAC assignments per §6.2
Config in git	.github/workflows/*.yaml, systemd units, quadlet on runtime (not secrets)

12. Monitoring and operations

Item	Consideration
Azure Monitor	Banking baseline: Layer 1 golden image installs rsyslog fragment + AMA prereqs; Layer 2 <code>monitor-baseline</code> module attaches DCR (syslog + perf) and <code>AzureMonitorLinuxAgent</code> to both VMs → client Log Analytics / Sentinel
Logic Monitor	Patrick—OS, disk, Podman, systemd failed units (separate from Azure Monitor; both may coexist)
Alerts	Disk full (ingestion), service down, high CPU
Log forwarding	Azure Monitor DCR → Log Analytics; SIEM rules client-owned
Runbooks	Restart service, disk full, failed deploy rollback
On-call	Client Linux admin / platform

13. DR and business continuity

Item	Consideration
ASR	Client plans ASR for Linux—include data disk and Podman volume paths
RPO/RTO	Define with business before prod
Rebuild	VM + cloud-init + pull images + restore data disk
Region	DR region replication if required

14. Licensing and support

Item	Owner
RHEL subscriptions	Client procurement
RHSM / Satellite	Register at build; network path to Red Hat
Azure compute cost	Finance chargeback tags

15. Build methods (layered model)

Industry-standard approach for this client: **base golden image (OS only)** → **role bootstrap** → **workload orchestration**. See ARCHITECTURE.md layering table.

Layer	What	Includes	Excludes
1 — Golden image (SIG/AIB)	Shared RHEL 9 base for runner + runtime VMs	CIS hardening, monitoring agent, <code>azure-cli</code> , <code>auditd</code> , SELinux, <code>chrony</code> , compliance artifacts	FastAPI, Dagster, ingestion services, Podman app containers, <code>actions-runner</code> , secrets, UAMI/RBAC assignment
2 — Role bootstrap	cloud-init / Ansible on first boot	Runner: <code>actions-runner</code> , build deps, UAMI attach. Runtime: Podman engine, data disk, Quadlet stack (<code>api</code> + <code>nginx</code> sidecar templates)	Running app containers, prod data
3 — Workload orchestration	GitHub Actions + <code>systemd</code>	Pull images, start Dagster/FastAPI/ingestion via <code>systemd/quadlet</code> , KV secrets	OS baseline changes

Path	When	Notes
A — IaC VM from marketplace + cloud-init/Ansible	Default if single/few VMs	Layers 2–3 without SIG; layer 1 = marketplace RHEL + Ansible hardening
B — IaC VM from SIG base + role bootstrap	Recommended when fleet or audit wants versioned OS	Terraform three-layer reference — see below
C — Clone dev VM	Never for UAT/prod	Carries shared admin, manual mounts, pen test debt

Three-layer Terraform (reference)

Layer	Apply path	Delivers
1	infra/terraform/environments/prod/ (layer1-image-pipeline)	SIG golden image: CIS, <code>azure-cli</code> , Python/pip/venv/Ansible , Azure Monitor prereqs + Logic Monitor stubs
2	infra/terraform/environments/workload/	Runner + runtime VMs, UAMI, monitor-baseline (DCR + AMA), data disk, optional file share; cloud-init runs layer-2 Ansible
3	GitHub Actions (client)	<code>podman pull</code> , systemd/quadlet services — <code>layer3_handoff</code> output in workload stack

Recommended sequence: Stabilize **dev** → **layer 1** SIG build → **layer 2** UAT VMs → **layer 3** CI validation → **prod**.

One SIG image, two VM roles: Same golden image for GitHub runner and Podman runtime; cloud-init calls `/opt/compliance/bootstrap/run-layer2-ansible.sh <role>`; UAMI RBAC per §6.2.

16. Pre-build checklist (UAT / prod)

Client platform completes before first boot (**both VMs**):

- ☐ Subscription and RG chosen (dev vs prod sub)
- ☐ **Two subnets/NSGs** (or one subnet with distinct NSGs): runtime vs runner
- ☐ Firewall allow-list: runtime vs runner egress (GitHub, registry, **Azure control plane** for UAMI)
- ☐ SKU sized **separately** from dev runtime vs runner metrics
- ☐ **Runtime:** OS + **managed data disk** (`data_disks`); fstab/cloud-init tested on UAT
- ☐ **Optional:** Azure File Share provisioned (`storage-fileshare`); NFS mount tested if shared ingestion
- ☐ **Runner:** disk sized for build cache; prune policy documented
- ☐ RHEL registered (RHSM) on both
- ☐ Entra SSH or per-user access model
- ☐ **Two UAMIs** created and attached (runner + runtime); RBAC per §6.2 (`AcrPush` / `AcrPull` , KV Secrets User)
- ☐ `azure-cli` installed on **both** VMs; `az login --identity` validated (ACR login smoke test)
- ☐ Runner UAMI: push to registry; runtime UAMI: pull only — **no** shared identity, **no** Contributor
- ☐ GitHub self-hosted runner registered; deploy workflow tested against UAT runtime
- ☐ **Runner removed** from legacy combined dev host (or decommission path documented)
- ☐ Pen test **open POs** addressed on both roles

- ☐ Azure Monitor: `log_analytics_workspace_id` in workload tfvars; DCR + AMA on both VMs; egress allow-list for `*.ods.opinsights.azure.com`
- ☐ Logic Monitor collector enabled (Patrick) if required alongside Azure Monitor
- ☐ Monitoring + alerts (runner: queue depth, disk; runtime: services, ingestion disk)
- ☐ ASR / backup (**runtime** includes data disk; runner often rebuild-from-IaC acceptable)
- ☐ Runbooks: deploy, rollback, runner disk full, failed job isolation

17. Consultant vs client

Activity	Vaco	Client
VM sizing / disk / Podman standards	Draft	Approve
Network requirements / allow-list	Draft	Implement
Terraform apply	Out of scope	Platform
Podman walkthrough + dev assessment	Lead	Anatoliy
Pen test remediation	Advise priority	Security executes
UAT/prod build	Support	Platform

18. Open inputs still needed

Input	From
Dev Podman metrics (CPU/RAM/disk)	Assessment on Linux VM
build → store → deploy walkthrough	Anatoliy
Pen test summary for Linux	Kirk / security
Azure Reader + diagram	Patrick / Anatoliy
Registry choice (ACR vs GHCR)	Client
UAMI + RBAC assignment model (scope per env)	Patrick / platform
APIM in front of VM or direct	Anatoliy — INGRESS-DECISION-NGINX-SIDECAR.md

Industry references

Full map: INDUSTRY-REFERENCES.md

Topic	Source
VM security	MCSB compute · Trusted Launch
Landing zone	Azure CAF Landing Zone
Identity on VMs	Managed identities
CI/CD runners	GitHub self-hosted runners
Containers	NIST SP 800-190 · Podman Quadlet
DR	Azure Site Recovery
OS baseline	CIS RHEL 9

Document history

Version	Date	Notes
0.1	<i>[date]</i>	Initial outline post-discovery
0.2	2026-05-28	Split GitHub runner VM from Podman runtime VM
0.3	2026-05-28	UAMI + Azure CLI + RBAC on both VMs (§6.2)